

Functional Annotation of *moaB-II* (Q88E67, PP_4600) in *Pseudomonas putida* KT2440

1. Identity verification (MANDATORY)

- **UniProt:** Q88E67 — 172 aa protein, *Pseudomonas putida* KT2440.
- **Gene:** `moaB-II`; ordered locus `PP_4600` (EMBL AAN70173.1).
- **Description:** Molybdenum cofactor biosynthesis protein B.
- **Family:** MoaB/Mog family. **Domains:** MoaB (IPR012245), MoaB/Mog-like domain superfamily (IPR036425), MoaB/Mog domain (IPR001453), MoaB_proteobac (IPR013484), MoCF_biosynth conserved site (IPR008284).
- **UniProt keywords:** Molybdenum cofactor biosynthesis; GTP-binding; Nucleotide-binding.

The gene symbol, organism, protein family and domain architecture retrieved from UniProt are **mutually consistent**: `moaB-II` denotes a *moaB* paralog ("copy II") of the molybdenum-cofactor (Moco) biosynthesis protein B in *P. putida*. Literature on the MoaB/Mog family (*E. coli* MoaB/MogA, archaeal MoaB, plant Cnx1, gephyrin) aligns with the domain/family assignment. **No literature describes a different, unrelated gene under this symbol that would create ambiguity.** The caveat is not mis-identification but *evidence type*: there is **no primary experimental study of the *P. putida* protein Q88E67 itself**; its annotation is by homology to characterized family members.

2. Summary answer

Q88E67/MoaB-II is a small (~19 kDa) cytoplasmic protein of the **MoaB/Mog family**, a paralog of MogA, predicted to participate in the terminal steps of **molybdenum-cofactor (Moco) / molybdopterin biosynthesis**. The MogA branch of this family catalyzes the **Mg²⁺/ATP-dependent adenylation of molybdopterin (MPT → MPT-AMP)**, the activated intermediate from which molybdenum is inserted (by MoeA) to complete Moco. However, **MoaB-type**

proteins — including Q88E67 by its UniProt annotation — bind molybdopterin (and GTP) but lack MPT-adenylyltransferase activity, and in *E. coli* MoaB is genetically dispensable for Moco synthesis. MoaB-II is therefore best described as a **MogA-like pterin/nucleotide-binding protein of the Moco pathway whose precise catalytic role is enigmatic and probably accessory/redundant**, acting in the cytoplasm.

3. Biochemical function and substrate specificity

3.1 The MoaB/Mog family and the reaction it centers on

The MoaB/Mog family unites bacterial MogA and MoaB, the plant multidomain protein Cnx1 (C-terminal G-domain), the animal protein gephyrin (G-domain), and *Drosophila* Cinnamon. Genetic complementation first showed that the C-terminal domain of Cnx1 "is homologous to the *E. coli* Moco proteins MoaB and MogA" and that these proteins act in "the last step of Moco biosynthesis, that is, the insertion of molybdenum into molybdopterin" (Stallmeyer et al., 1995,

P 8528286

Structural work confirmed all family members are "involved in the insertion of an unknown molybdenum species into molybdopterin to form the molybdenum cofactor" (Bader et al., 2004,

P 15159566

Mechanistically, the reaction proceeds in two steps: the **MogA/G-domain** first **adenylylates molybdopterin**, and **MoeA/E-domain** then uses this MPT-AMP intermediate plus molybdate to insert Mo. Bevers et al. (2008) demonstrated directly that the MogA-type activity "catalyzes the adenylation of MPT in a Mg^{2+} and ATP-dependent way"

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Structural work on the eukaryotic G-domains confirms the chemistry and the intermediate: the plant Cnx1 G domain produces "adenylated molybdopterin (molybdopterin-AMP), a mechanistically important intermediate... subsequently processed in a magnesium-dependent reaction by the amino-terminal E domain of Cnx1 to yield active molybdenum cofactor" (Kuper et al., 2004,

P 15306815

and in human gephyrin the "N-terminal G domain (GephG)" catalyzes "the penultimate adenylation of molybdopterin (MPT)" while the E domain performs "deadenylation as well as molybdate binding and insertion" (Kasaragod & Schindelin, 2016,

P 27112598

These G domains are homologous to *E. coli* MogA (Schwarz et al., 2001,

P 11554796

— the paralog family to which MoaB-II belongs. In *P. putida* the corresponding molybdate-insertion enzyme is MoeA (PP_2123), annotated to catalyze "the insertion of molybdate into adenylylated molybdopterin with the concomitant release of AMP."

3.2 What distinguishes MoaB from MogA — and what Q88E67 is annotated to do

Although MoaB and MogA are close structural paralogs, they are **functionally non-equivalent**. In the same study, "only MogA is able to catalyze MPT adenylylation, whereas *E. coli* MoaB is inactive" (Bevers et al., 2008,

P 18154309

The UniProt functional annotation of Q88E67 mirrors this precisely:

"May be involved in the biosynthesis of molybdopterin. Can bind GTP and has low GTPase activity. Can bind MPT, but has no MPT adenylyl transferase activity." (UniProt Q88E67, ARBA ECO:0000256)

Substrate/ligand specificity (by annotation transfer): binds **molybdopterin (MPT)** and **GTP** (nucleotide-binding, GTP-binding keywords), with only **low GTPase activity** and **no adenylyltransferase catalysis**. It therefore appears to retain the pterin/nucleotide-binding pocket of the MogA scaffold without performing the defining MogA chemistry.

3.3 Genetic evidence for an accessory/redundant role

In *E. coli*, the physiological role of MoaB has remained "enigmatic": transposon insertions in *moaB* caused partial Moco-deficiency (likely polar effects on the *moaABCDE* operon), but "in-frame deletions of *moaB*, or *moaB* overexpression, had no effect on either phenotype," data "consistent with the lack of any role for MoaB in Moco biosynthesis in *E. coli*" (Kozmin &

Schaaper, 2013, [P 23680484](#)).

This supports interpreting MoaB-II as **dispensable/redundant or specialized** rather than catalyzing an essential, unique step. The "-II" designation indicates *P. putida* encodes more than one MoaB-type protein, consistent with paralog redundancy or subfunctionalization.

3.4 Evolutionary/genomic placement in *P. putida* (this work — bioinformatic analysis)

Enumerating the MoaB/Mog family in the *P. putida* KT2440 proteome (UniProt, taxon 160488) returns **exactly two members: moaB-I** (Q88L15, 179 aa, PP_2122) and **moaB-II** (Q88E67, 172 aa, PP_4600). Notably, the proteome contains **no gene annotated as MogA / molybdopterin adenylyltransferase** (0 hits), although it does encode the downstream **MoeA** (Q88L14, PP_2123) and **MobA** (Q88HA3, molybdenum cofactor guanylyltransferase).

Pairwise global alignments (Needleman–Wunsch, identity scoring) place moaB-II firmly in the **MoaB clade, not the MogA clade**:

Comparison	% identity
MoaB-II (Q88E67) vs <i>E. coli</i> MoaB (P0AEZ9)	55.2%
MoaB-II (Q88E67) vs <i>E. coli</i> MogA (P0AF03)	27.7%
MoaB-II vs MoaB-I (Q88L15)	72.6%
<i>E. coli</i> MoaB vs MogA (reference)	30.3%

Because *E. coli* MoaB is the enzymatically **inactive** paralog and MoaB-II is ~55% identical to it (but only ~28% to the active MogA), sequence evolution independently supports the annotation that MoaB-II **lacks MogA-type adenylyltransferase activity**. MoaB-I and MoaB-II are 73% identical to each other, indicating a **gene-duplication** origin. Genomic context reinforces a division of labour: **moaB-I (PP_2122) is immediately adjacent to moeA (PP_2123)** — the two forming a syntenic Mo-insertion module — whereas **moaB-II (PP_4600) sits at a separate, distant locus**, most parsimoniously an accessory/duplicate copy.

4. Structure

MoaB proteins adopt a **modified Rossmann fold** and oligomerize. The 1.6 Å *E. coli* MoaB structure showed the protein is "assembled into a hexamer composed of two trimers" whose monomers have "high structural similarity with two proteins, MogA and MoeA," and with the G-domains of gephyrin and Cnx1; a putative active site was identified (Sanishvili et al., 2004,

P 15269205

A second structure described a "32-symmetric hexamer" with a fold "similar to those of the MogA protein of *E. coli*, the G-domains of rat and human gephyrin and the G-domains of Cnx1 protein from *A. thaliana*" (Bader et al., 2004,

P 15159566

Archaeal MoaB (ST2315) likewise forms trimers/hexamers with a fold "similar to those of other enzymes... involved in the molybdopterin- and molybdenum cofactor-biosynthesis pathways" (Antonyuk et al., 2009,

P 20054111

At 172 aa, Q88E67 is the size of a single MogA/G-type domain and is expected to fold identically and present the conserved pterin-binding pocket, most likely assembling into a trimer/hexamer. This is **structural/bioinformatic evidence** supporting the pathway assignment.

5. Localization

Moco biosynthesis is a **cytoplasmic** process, and the soluble MoaB/Mog-family proteins act there. Consequently MoaB-II is predicted to function in the **cytoplasm**, where it can access molybdopterin/MPT intermediates and nucleotides. No experimental localization for Q88E67 is available; the assignment follows from the pathway's known compartment and the protein's soluble, cofactor-binding nature.

A sequence-topology analysis performed here corroborates a soluble cytoplasmic protein: Kyte–Doolittle hydropathy (window 19) peaks at only 1.47 with **no window exceeding the ~1.6–1.8 transmembrane threshold** (zero predicted TM helices); the N-terminus lacks a cleavable Sec/Tat signal-peptide architecture (its hydrophobic start is the first β-strand of the

Rossmann fold); and the protein has balanced charge (19 Asp+Glu vs 18 Lys+Arg) and near-neutral GRAVY (+0.137) — hallmarks of a soluble globular cytoplasmic enzyme rather than a membrane or secreted protein.

6. Pathway context and biological importance in *P. putida*

MoaB-II sits within **cofactor biosynthesis → molybdopterin/Moco biosynthesis** (UniProt PATHWAY annotation). This pathway is metabolically important in *P. putida*, which deploys numerous **Moco-dependent molybdenum hydroxylases**. Examples include the molybdopterin-cytosine-dinucleotide (MCD)-dependent enzymes of nicotine and N-heterocycle catabolism — where "a novel gene, PPS_4397 encoding moaE, is necessary for molybdopterin cytosine dinucleotide biosynthesis" (Jiang et al., 2015,

[P 26304596](#)

and the 3-succinoylpyridine step uses "molybdenum molybdopterin cytosine dinucleotide as a cofactor" (Tang et al., 2013,

[P 24204321](#)

— as well as quinoline/isoquinoline 2-oxidoreductases expressed in KT2440 (

[P 12654012](#)

[P 12023088](#)

[P 12730200](#)

and heterologous xanthine oxidase (

[P 41933999](#)

The Moco pathway that MoaB-family proteins support therefore underpins *P. putida*'s catabolic versatility toward aromatic and N-heterocyclic compounds and purine oxidation. **The report deliberately does not attribute broad pleiotropic phenotypes to MoaB-II specifically**, given the *E. coli* evidence that MoaB deletion is phenotypically silent.

7. Supported vs. open/refuted statements

Supported (by homology + family biochemistry/structure): - Q88E67 belongs to the MoaB/Mog family and acts in the cytoplasmic Moco/molybdopterin biosynthetic pathway. - It binds molybdopterin and GTP and adopts a MogA-like G-domain fold (likely oligomeric). - It does **not** possess the MogA MPT-adenylyltransferase activity (per its annotation and *E. coli* MoaB biochemistry).

- MoaB-II belongs evolutionarily to the (inactive) proteobacterial MoaB clade, not the MogA clade (~55% vs ~28% identity), and is one of two duplicated MoaB paralogs; moaB-I is syntenic with moeA while moaB-II is at a separate locus.

Open / not established: - The exact physiological reaction (if any) uniquely catalyzed by MoaB-II. In *E. coli*, MoaB is dispensable; whether *P. putida* MoaB-II is redundant, conditionally required, or subfunctionalized (e.g., under specific Mo/W or substrate conditions) is untested. - *P. putida* KT2440 encodes **no canonical MogA**, so which protein supplies the adenylated-MPT (MPT-AMP) substrate that MoeA consumes is unresolved — candidates are latent activity of a MoaB paralog or a molybdate-concentration-dependent bypass (MogA is non-essential in *E. coli*). - Direct experimental data on Q88E67 (activity assays, structure, deletion phenotype, localization) — none exist; all statements are inferences.

8. Limitations and future directions

- **No primary study of Q88E67.** All functional statements rest on annotation transfer from *E. coli* MogA/MoaB, archaeal MoaB, and eukaryotic Cnx1/gephyrin, plus automated (ARBA/PIRNR) UniProt annotation.
- **Family functional ambiguity.** Even in *E. coli* the role of MoaB is unresolved; this uncertainty propagates to MoaB-II.
- **Recommended experiments:** (i) construct in-frame PP_4600 deletion and test Moco-dependent phenotypes (nitrate reductase/chlorate resistance, molybdoenzyme activities) alone and in combination with the other *P. putida* moaB/mogA paralog(s); (ii) in vitro assays for MPT adenylyltransferase and GTPase activity of purified Q88E67; (iii) solve the structure / AlphaFold-model the pterin/nucleotide pocket and compare to MogA vs. *E. coli* MoaB to identify residues explaining loss of adenylyltransferase activity; (iv) test conditional requirement under Mo limitation or tungsten substitution.

Key references

- Bevers et al., 2008, *Function of MoaB proteins in the biosynthesis of the molybdenum and tungsten cofactors*
([P 18154309](#)).
- Kozmin & Schaaper, 2013, *Genetic characterization of moaB mutants of E. coli*
([P 23680484](#)).
- Sanishvili et al., 2004, *Crystal structure of E. coli MoaB*
([P 15269205](#)).
- Bader et al., 2004, *Structure of MoaB of E. coli*
([P 15159566](#)).
- Stallmeyer et al., 1995, *A. thaliana cnx1...*
([P 8528286](#)).
- Antonyuk et al., 2009, *Structure of Mo-cofactor biosynthesis protein B (ST2315)*
([P 20054111](#)).
- Kuper et al., 2004, *Structure of the molybdopterin-bound Cnx1G domain*
([P 15306815](#)).
- Kasaragod & Schindelin, 2016, *Structural framework for metal incorporation during Moco biosynthesis*
([P 27112598](#)).
- Schwarz et al., 2001, *Crystal structures of human gephyrin and plant Cnx1 G domains*
([P 11554796](#)).
- P. putida molybdoenzyme context:
([P 26304596](#))
([P 24204321](#))

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P 41933999

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